



Comparative analysis of single-view and multiple-view data collection strategies for detecting partially-occluded grape bunches: Field trials

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ABSTRACT

Extracting phenotypic traits of grape bunch is crucial for accurately monitoring grape quality, health, and yield estimation. This is important for optimising resources, enhancing marketing strategies, and boosting overall agricultural productivity. While most research concentrates on data processing algorithms, this study focused on the preceding step: collecting reliable data. Object detection and tracking enable precise monitoring and quantification of fruit, facilitating agricultural management. This study compares two data acquisition methodologies for grape bunch detection and tracking in a commercial vineyard where leaf removal was not performed: a traditional single-view approach and a multiple-viewing method designed to mitigate fruit occlusion issues. The PointTrack algorithm, trained and validated using MOTSA annotations, was employed to evaluate detection and tracking performance through metrics of three trials. The multiple-view method achieved i) higher ratio between tracked and GT detections of 74 % compared to 23 % for the single-view approach and ii) enhanced tracking metrics, with the multiple viewing trials metrics ranging from -1.35 to 3.84 for MOTSA (Multiple Object Tracking and Segmentation Accuracy) and sMOTSA (soft MOTSA), and iii) higher correlation and lower RMSE of grape bunch phenotypic traits (OIV codes 202 and 203) compared to ground truth measurements ($R^2 = 0.53$, $RMSE = 19.13$). Nonetheless, the multi-view technique was compromised by motion blur due to UAV movements, complicating the tracking process. This study underscores the importance of strategic data acquisition in improving performance for fruit detection and tracking. Future work should extend this methodology to other fruit varieties and environments to validate its broader applicability, enhancing the reliability of yield estimation in precision agriculture.

1. Introduction

Deep Learning (DL) has been applied in recent years in the agricultural domain, for instance, in fruit detection and tracking [1,2], witnessing notable advancements and offering promising research lines for yield prediction [3–5] and fruit growth monitoring [6,7], among others. Within DL, Convolutional Neural Networks (CNNs) have proven to be particularly effective since they learn complex features from images and videos, which allow accurate fruit recognition in complex field environments, such as challenging illumination conditions [8]. These technologies offer great potential in the agricultural sector for fruit counting, phenotyping, and consequent yield prediction, abating inspection time and reducing labour costs.

One of the advantages of Deep Learning is domain adaptation [9,10], which consists of applying the knowledge gained in one specific field to another domain. For instance Ref. [11], developed a fruit flower detection for multiple species trained on apple flower images, and tested on apple, peach, and pear flowers. Further [12], implemented Faster R-CNN for multi-class fruit detection, including apple, mango, and orange labels, which benefits from a common characteristic between those fruits: their circularity.

Despite the potential of DL in fruit counting [13], challenges can lead to under or overestimation, for instance, occlusion [14,15]. Occlusion occurs when fruits are partially or fully hidden, typically by leaves and branches, complicating the detection and tracking task [16]. Fully hidden fruits, enclosed by leaves, are not detectable from any perspective,

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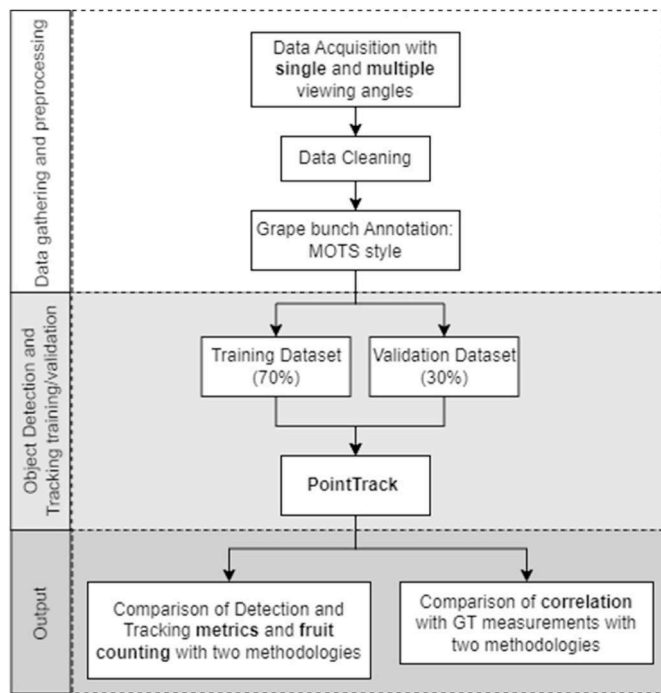


Fig. 1. Flowchart of this research. Data gathering was performed using two methodologies: single-view without any enhancement, and multiple-view to avoid occlusion. The next step was to carry out Data Cleaning and Data Annotation with the Multiple-Object Tracking and Segmentation (MOTS) style, followed by training and validating the object detection and tracking algorithm. Finally, a comparison of metrics, fruit counting, and correlation with GT measurements was executed between both methodologies.

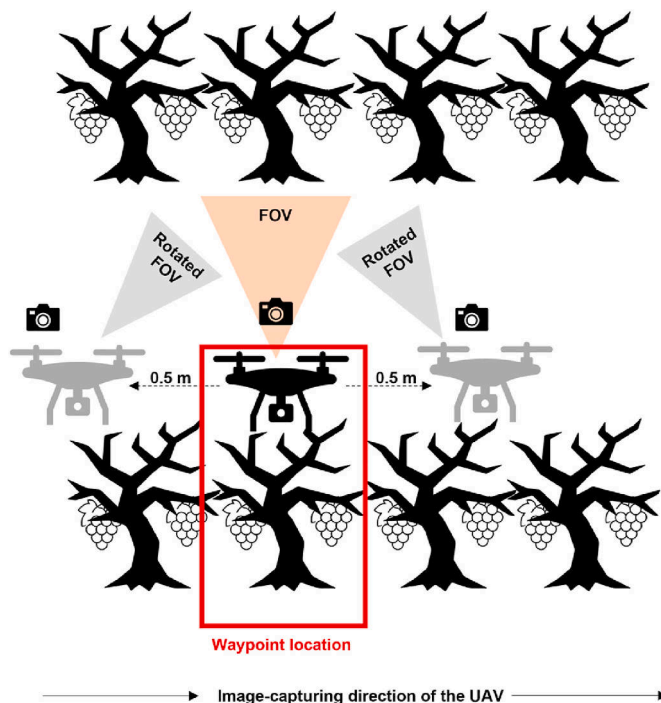


Fig. 2. Multiple viewing approach that collects images from three different perspectives. Adapted from Ref. [25].

whereas partially occluded fruits can be seen better from specific views [17]. proposed adding spherical shapes prior to detection, while [18] suggested a 3D fruit localisation method that determines centroid

coordinates and approximates shape. However, these solutions were only tested on spherical fruits, which does not apply to grape bunches [19]. proposed a method to detect ripe fruits by removing the complex background of colorful ripe fruits. However, this method is not effective when applied to grape bunches at an early stage of development or for white varieties.

Furthermore, detection and tracking metrics suffer from fruit occlusion. Several authors have focused on solving this reduction in metrics by developing post-processing techniques for pre-recorded datasets [20–22]. Nevertheless, the focus can also point to the optimal acquisition of the datasets by considering multi-view perceptions [23] and the specific knowledge domain for which they are being acquired [24]. Therefore, while most research targets data processing algorithms [18, 19], this study targets the initial step of gathering accurate data.

To detect partially occluded fruits, multi-view strategies have increasingly gained attention in agricultural research since their potential to enhance object detection and tracking has been shown. For example [25], developed a theoretical framework for UAVs to acquire images or videos from the side of the canopy considering multiple views (frontal, left side, and right side) to increase the chance of observing the fruit at its full size. [26,27], have further demonstrated the advantages of multi-view approaches for improving the detection and counting of fruits and flowers.

Following this rationale, this study validates the theoretical framework proposed by Ref. [25] in a real-world experiment conducted in a commercial vineyard with leaf occlusion. Unlike previous work, which primarily focused on theoretical modeling, this study provides an empirical assessment by comparing the results with those obtained using a traditional single-view methodology for data acquisition. It analyses key metrics such as grape bunch detection and tracking accuracy, fruit counting, and correlation with ground truth (GT) measurements. This study makes a unique contribution by contrasting the outcomes when fruit detection is performed with regard to the data acquisition method versus actively optimising data collection through the use of multiple views. By bridging the gap between theoretical modelling and practical application, the findings not only validate the proposed framework but also offer insights for improving fruit detection and counting in commercial agricultural settings.

2. Materials and methods

The research process that was carried out is outlined in Fig. 1, illustrating the employed workflow. The focus of attention was driven already to the first step: data acquisition. Two different methodologies were employed to gather the necessary videos in MP4 format. The first method used the traditional method that consisted of flying the UAV to record videos of the side of the canopy without considering any enhancement, similar to Ref. [28]. The second implemented a multi-view viewing approach (Fig. 2), which involves collecting data from various perspectives to reduce errors in fruit detection caused by leaves and branches blocking the view. It was designed specifically for woody crop environments, such as taking into account the biophysical characteristics of the field.

For both methods, the UAV used was a DJI Phantom 4 RTK (DJI Sciences and Technologies Ltd., Shenzhen, Guangdong, China) that has the sensor integrated. Flights were conducted at an altitude of 3 m above ground level (AGL), and the speed was not stable, ranging between 0.1 and 1 m/s. Videos were captured between September 19 and September 20, 2023, under sunny conditions with wind speeds below 0.5 m/s. This experimental setup aimed to test the hypothesis that the multi-view approach is more effective for object detection and tracking while addressing occlusion issues. For more details on the experimental setup, refer to Ref. [29].

The following steps, in white, were common for both methodologies and were composed of cleaning the videos, annotation of the grape bunches, and training/validating the PointTrack [30] algorithm. A total

Table 1

Grape bunch detection and tracking metrics (sMOTSA, MOTSA, MODSA, AND MODSP) of the three executed trials.

TRIAL 1					
Video	Category	sMOTSA	MOTSA	MODSA	MODSP
1	Single view	−1.18	−1.18	−1.18	100
2	Single view	−0.08	−0.08	−0.08	100
3	Multiple views	−16.21	−12.92	−11.08	64.18
4	Multiple views	−4.14	−3.76	−3.61	90.26
5	Multiple views	−2.62	−0.76	−0.28	71.17
TRIAL 2					
Video	Category	sMOTSA	MOTSA	MODSA	MODSP
1	Single view	−1.18	−1.18	−1.18	100
2	Single view	−0.08	−0.08	−0.08	100
3	Multiple views	−5.74	−5.5	−5.42	92.74
4	Multiple views	−2.06	−1.53	−1.42	85.5
5	Multiple views	−2.2	−0.8	−0.44	71.16
TRIAL 3					
Video	Category	sMOTSA	MOTSA	MODSA	MODSP
1	Single view	−7.51	−5.44	−4.98	67.92
2	Single view	−0.9	−0.48	−0.46	93.16
3	Multiple views	−7.08	−4.7	−4.14	65.93
4	Multiple views	−1.35	0.3	0.68	75.85
5	Multiple views	0.69	3.44	3.84	70.8

of 5958 frames were manually annotated in PNG format and include txt files with the grape bunch label. The workflow finalised with the comparison in metrics, fruit counting, and correlation with ground truth measurements for phenotyping obtained with each procedure. The dataset used in this study [29] is publicly available. It describes the data acquisition method and contains information about several UAV flights performed with a customised route planner that allowed data to be recorded using single-view and multiple-view approaches. It also includes the grape bunch annotations and the labelling process using the CVAT software [31].

2.1. PointTrack implementation and metrics

The implementation of the PointTrack algorithm in this study was largely similar to that in Ariza-Sentís et al. [32], except for image resolution differences due to the utilisation of different sensors. Transfer

learning [33] was employed to train the Spatial Embeddings (SE) model. An Adam optimiser [34] with a learning rate of 5×10^{-5} and a finetuning rate of 5×10^{-6} was used. The SE model [35] was trained with a batch size of 20 and 1200 epochs. Hyperparameters were optimized by adjusting their values up and down until no further improvement in performance or overfitting was detected. During SE training, image crops were vertically oriented to match the grape bunches' shape, with resolutions of 256 x 512 and 512 x 1024 pixels for initial training and finetuning, respectively. The PointTrack tracking component was trained with a batch size of 32 and 100 epochs.

Several metrics were computed to assess their ability to detect and track grape bunches to compare the two proposed methodologies and determine the most valid method for object detection and tracking in vineyards without leaf removal. Two detection performance metrics, MOTSP (Multiple Object Tracking and Segmentation Precision) and MODSP (Multiple Object Detection and Segmentation Precision) were calculated using Equations (1) and (2). Additionally, MOTSA (Multiple Object Tracking and Segmentation Accuracy) and sMOTSA (soft MOTSA) were obtained as detection metrics by applying Equations (3) and (4). The higher the metrics the more accurate on performance.

$$MOTSP = \frac{\tilde{TP}}{|TP|} \quad (1)$$

$$MODSA = \frac{TP}{|TP|} \quad (2)$$

$$MOTSA = \frac{|TP| - |FP| - |IDS|}{|M|} \quad (3)$$

$$sMOTSA = \frac{\tilde{TP} - |FP| - |IDS|}{|M|} \quad (4)$$

Where TP are true positives, which represent the number of masks that were properly detected compared to the ground truth masks, with an Intersection over Union (IoU) value over 0.5; $\tilde{TP}$ are soft true positives, which consists of the sum of IoU for all the TP; FP are false positives, which represent the number of masks identified as grape bunches when they actually were not; IDs which consist on the number of masks that switched ID compared to the previous frame in which they appeared; and M stands for the number of GT masks.

Lastly, a comparison regarding the total count of grape bunches per

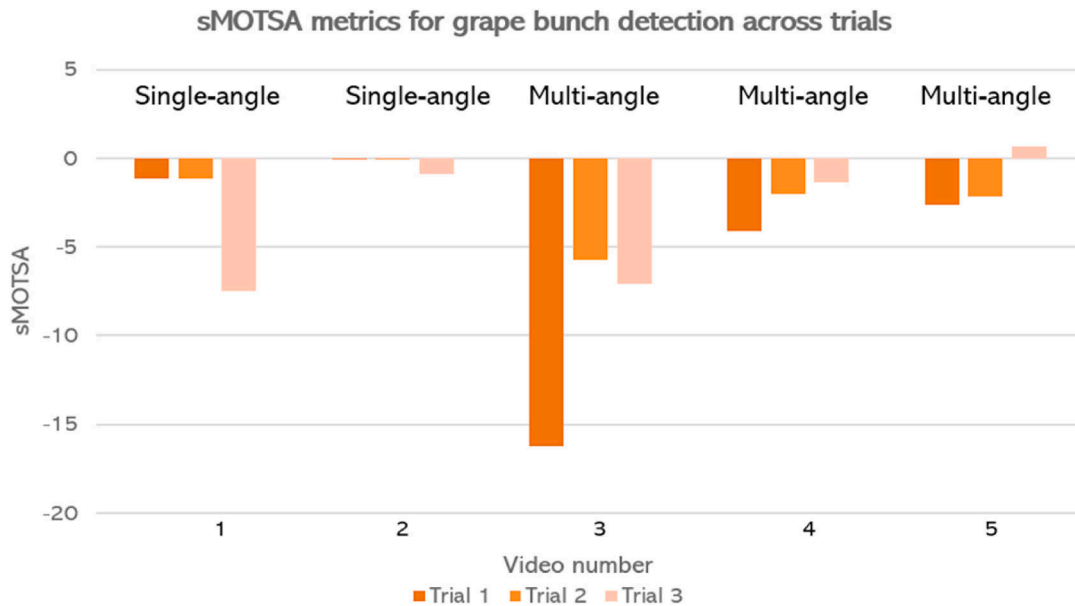


Fig. 3. Comparison of the sMOTSA metrics for grape bunch detection across trials.

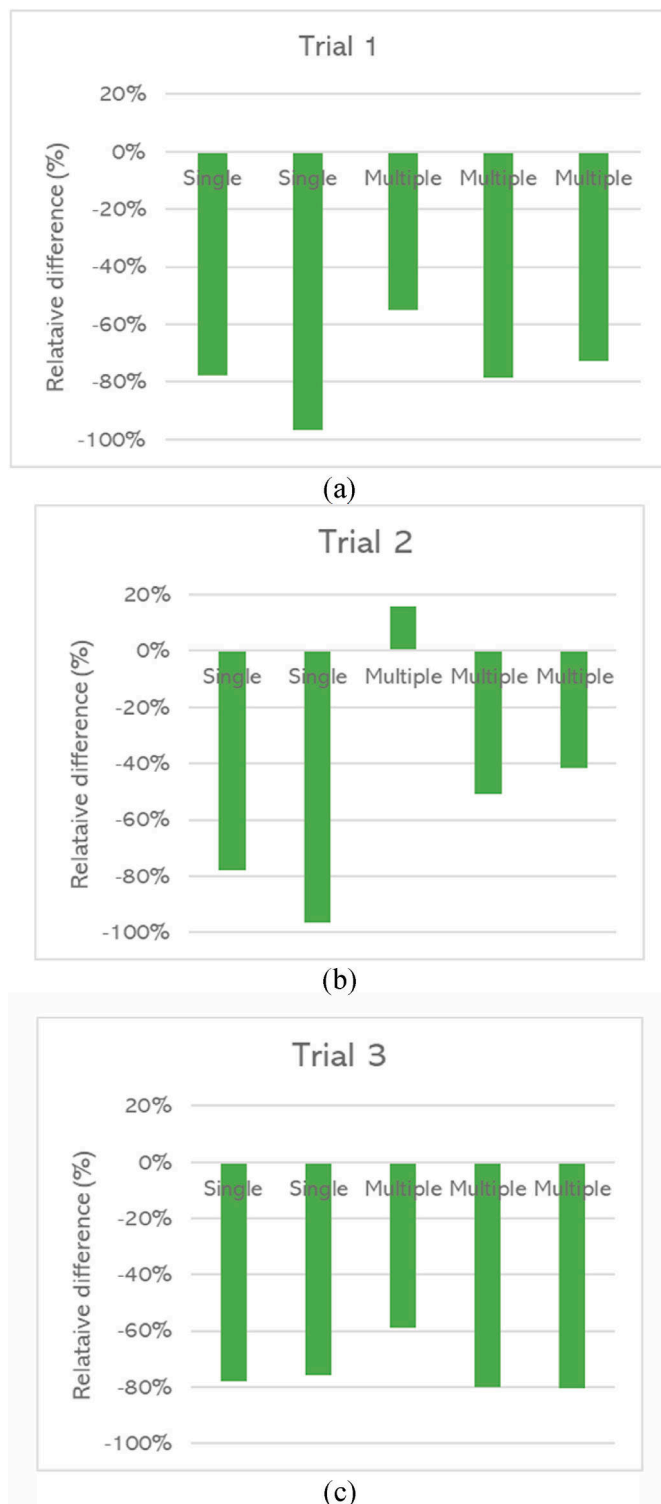


Fig. 4. Relative difference between the actual and predicted number of grape bunches for each trial. The values are negative due to undercounting.

video between the two methods and the real value obtained from the annotations was calculated, along with a correlation with ground truth measurements for phenotypic purposes.

2.2. Grape bunch measurements

The International Organization of Vine and Wine (OIV) [36] has established several standards for characterising the phenotyping of

grape bunches. For example, bunch length and width correspond to the descriptor codes OIV 202 and 203, respectively. According to the OIV guidelines, a sample size of 10 bunches is required to determine these descriptors. Therefore, this study analysed ten bunches to compare the phenotyping traits (grape bunch length and width) between annotated and detected bunches. Measuring these descriptors is crucial for gaining insights into the physical dimensions of the bunches, which serve as indicators of fruit quality and vine health [37]. Moreover, the size of grape bunches is directly related to the harvest yield, thus enabling more precise yield estimations [38].

3. Results

A series of experiments were conducted to evaluate the performance of the algorithm. The most effective trials were as follows.

- Trial 1: training the algorithm exclusively with multiple view data
- Trial 2: training the algorithm exclusively with single-view data
- Trial 3: training the algorithm with a combination of multiple and single view data

3.1. Detection and tracking metrics

Table 1 presents the detection and tracking metrics for each trial. The video number remained consistent across trials. Some values are negative due to the occurrence of false positives. It was observed that for the single-view dataset, Trial 2 achieved the highest detection and tracking metrics (sMOTSA, MOTSA and MODSA equal to -0.08), while for the multiple-view dataset, Trial 3 exhibited the highest metrics (positive values for all sMOTSA, MOTSA and MODSA). This can be attributed to the fact that Trial 2 included only single-view data, allowing the algorithm to learn the patterns of that specific dataset effectively. In contrast, Trial 3, which incorporated both multiple and single-view data, enabled the algorithm to generalise better for the multiple-view scenario.

Fig. 3 provides a visual comparison of the sMOTSA metrics across trials for each category (single and multiple angle).

3.2. Grape bunch counting

In addition to the detection and tracking metrics, the algorithm's ability to count the number of grape bunches was evaluated. Fig. 4 shows the relative difference between the ground truth number of grape bunches present in each video and the predicted number. In most cases, the value is negative, indicating undercounting. The closer the value is to 0, the more accurate the detection count. Positive values indicate overcounting of the predicted grape bunches.

3.3. Single and multiple views to avoid occlusion

Furthermore, to visually validate the importance of occlusion, several images of the same plant, both with single and multiple views, are presented in Fig. 5. It can be observed that incorporating multiple points of view reduces the occlusion of grape bunches, making them easier to detect and track.

3.4. Phenotypic assessment

Table 2 displays the annotated and detected masks of 10 grape bunches for both single-view and multiple-view perspectives. It can be observed that the sizes and shapes differ between single and multiple views, even for the same grape bunch GT, due to fruit occlusion, as highlighted in Fig. 5, which plays a critical role in hiding full shapes. In addition, grape bunches appear more accurately shaped in the multiple-view masks since they are completely visible from at least one of the three views. When the grape bunch is fully observed from all three



Fig. 5. Comparison of the same plant seen with a single view (with occlusion) and multiple views (without occlusion) dataset. The red circles indicate the location of the occluded grape bunches, which are not visible from a single view. The different colours of the grape bunches on the right column indicate different detections.

views, the best one, which is less blurred, with a lower percentage of occlusion, and with a better contrast with the surrounding canopy, is selected for extracting their phenotypic traits.

Further, Fig. 6 compares the ground truth measurements of each bunch's dimensions (length and width, in pixels) and the predicted measurements obtained from single and multiple perspectives. It can be observed that the methodology proposed had a higher correlation and lower RMSE ($R^2 = 0.53$, $RMSE = 19.13$) compared to single-view ($R^2 = 0.39$, $RMSE = 28.05$).

4. Discussion

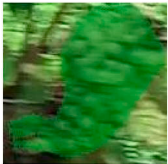
Fruit occlusion presents a significant challenge in object detection and tracking [14,39,40]. A multiple-view method was implemented to address this issue, capturing images from various viewpoints to reduce occlusion and enhance visibility. Although the detection and tracking metrics were not the primary focus, the observed limitations, such as false positives due to colour similarities and varying illumination

conditions [41,42], were similar to the problem presented by Ref. [32].

Despite these challenges, the multiple-view approach demonstrated a more accurate count of grape bunches compared to single-view methods. Fig. 4 showed that the relative difference between the actual and predicted number of grape bunches was more accurate with multiple views, despite significant undercounting in most cases. This can be expected because of occlusions, making underestimation more likely than overestimation, except in cases where there are a lot of FPs. The ratio between Tracked and GT values for single view ranged from 13 % to 23 %, while multiple views consistently outperformed, ranging from 27 % to 74 %, with the highest metrics observed in the second trial.

Illumination conditions varied across trials and datasets, necessitating validation to ensure shadows did not significantly impact detection. Table 2 remarks on the shape differences between single-view and multiple-view detections, with the latter providing a more realistic representation of grape bunches. This improvement in morphology, as shown in Table 2, highlights the ability of multiple views to better capture the shape of grape bunches, aligning them more closely with

Table 2
Grape bunch annotations (second and fourth columns) and bunch mask detections obtained with PointTrack (third and fifth columns). The shapes of the same grape bunch number differ from a single and multiple-view viewing due to occlusion.

Grape bunch number	GT single-view	Single-view detection	GT multiple-view	Multiple-view detection
1				
2				
3				
4				
5				
6				
7				

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Table 2 (continued)

Grape bunch number	GT single-view	Single-view detection	GT multiple-view	Multiple-view detection
8				
9				
10				

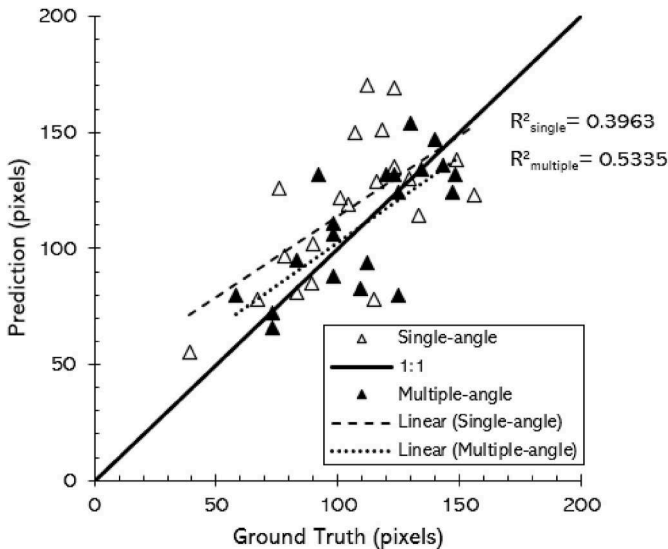


Fig. 6. Correlation between the ground truth measurements of the grape bunch dimensions (OIV 202:length and OIV 203:width, in pixels) and the predicted measurements using single-view and multiple-view methodologies.

ground truth shapes. Fig. 5 further supported the effectiveness of the multiple-view method, showing increased correlation with ground truth measurements from an R^2 of 0.39 with single-view to 0.53 with multiple-view, and a reduction in RMSE from 28.05 to 19.13. Additionally, Fig. 5 illustrates that using a single perspective results in missed detections, while multiple perspectives mitigate this issue. However, tracking using three views presented difficulties due to motion blur from UAV movement, causing the camera to lose track of the fruit, and impacting the continuity of ID numbers through frames. This issue was particularly evident in grape bunches 5, 6, and 7.

4.1. Limitations and future research

Several limitations were encountered during the development of the research. One significant challenge was the difficulty in labelling the

videos due to the homogeneity between white grape bunches and the surrounding green canopy. This issue was further complicated by challenging illumination conditions, which often made it hard to distinguish between fruit and shadows. Moreover, the process of annotating videos captured from multiple angles presented difficulties in identifying the same grape bunch across different frames. Despite these obstacles, the labelling was successfully completed, demonstrating resilience to the inherent environmental variability encountered in agricultural settings.

No specific limitations were observed regarding the selected UAV, which was chosen for its affordability and suitability for farmers. While more advanced sensors could have been employed, an integrated sensor was selected to enhance reproducibility. The UAV and sensor setup proved sufficient accuracy for the objectives of this study.

During the grape bunch annotation process, some challenges were found. Continuity in tracking annotated grape bunches, especially when they disappeared for several frames, was complicated but crucial for maintaining accurate metrics. We followed a similar annotation procedure as described in Ref. [43], ensuring that grape bunches reappearing in subsequent frames retained the same ID. While some datasets, allow for discontinuous tracks [44], we found that this approach led to lower tracking metrics in our study. Maintaining continuity proved essential for accurate detection and tracking in our MOT-based framework.

Future research should include vineyards from different latitudes and varieties to train the object detection and tracking algorithm in other environments. Expanding the dataset to include diverse latitudes would enhance the algorithm’s robustness and generalisation. Although the current study utilised a significant volume of video data, incorporating additional variations would provide further improvements. Furthermore, the methodology could be extended to red grape varieties and other fruit species to validate the results and broaden the scope of application.

The replication of experiments across single and multiple perspectives validated the added value of having multi-view strategies in commercial vineyard conditions, further supporting the theoretical framework. Despite these challenges, the improvement achieved by including multiple views over a single view emphasises the importance of a well-structured data acquisition plan. Proper planning can significantly enhance the accuracy of fruit detection and tracking, even under non-ideal conditions. This study demonstrates that multiple views are a

valuable strategy for improving the detection and assessment of partially occluded fruit, which is essential for several operations such as accurate phenotyping, reliable yield estimation or disease detection. Future work should implement the same methodology in red varieties and other fruit species to confirm the obtained results.

5. Conclusions

This study demonstrates that employing multiple perspectives for fruit detection and tracking is more effective than relying on a single perspective, as it minimises occlusion and enhances the robustness of the algorithm. In addition, it highlights the significance of strategic data acquisition in improving deep learning models for object detection and tracking. The PointTrack algorithm was utilised to gather detection and tracking metrics, along with assessing phenotypic traits in grape dimensions (OIV codes 202 and 203), which were then compared to ground truth values. Using a multi-view approach significantly reduced occlusion, enhancing the visibility and counting accuracy of grape bunches compared to conventional single-view methods, leading to more precise phenotyping and, consequently, better yield predictions. The multi-view approach achieved a ratio between tracked and GT detections of 74 %, while the single-view method only identified 23 % of the present grape bunches. Moreover, incorporating multiple perspectives improved the correlation with ground truth measurements, increasing the R^2 from 0.39 with a single view to 0.53 with multiple views, and reducing the RMSE from 28.05 to 19.13. Future research should expand this methodology to other types of fruit and settings to confirm its wider applicability.

CRedit authorship contribution statement

Mar Ariza-Sentís: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hilmy Baja:** Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Sergio Vélez:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Methodology, Investigation, Conceptualization. **Rick van Essen:** Writing – review & editing, Software, Methodology, Investigation, Formal analysis, Data curation. **João Valente:** Writing – review & editing, Validation, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The dataset has been uploaded and mentioned in the manuscript

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